1 I

1.1 (1)

The equation of motion for an electron is given by the Lorentz force:

$$m\frac{d\vec{v}}{dt} = -e\left(\vec{E} + \vec{v} \times \vec{B}\right).$$

1.2 (2)

Assuming that the electron velocity oscillates with the same angular frequency as the electromagnetic wave, we write

$$\vec{v} = \vec{v}_0 e^{-i\omega t}.$$

Substituting this into the equation of motion gives

$$-i\omega m\vec{v} = -e\left(\vec{E} + \vec{v} \times \vec{B}\right).$$

Since $|\vec{B}| \simeq |\vec{E}|/c$ and $v \ll c$, the magnetic contribution is negligible compared with the electric contribution:

$$|\vec{v} \times \vec{B}| \lesssim \frac{v}{c}|\vec{E}| \ll |\vec{E}|.$$

Therefore,

$$-i\omega m\vec{v} \simeq -e\vec{E},$$

which gives

$$\boxed{\vec{v} = -\frac{ie}{m\omega}\vec{E}}.$$

1.3 (3)

The current density produced by the motion of the electrons is

$$\vec{j} = -ne\vec{v}.$$

Using the result from (2), we obtain

$$\boxed{\vec{j} = \frac{ine^2}{m\omega}\vec{E}}.$$

1.4 (4)

Starting from Faraday's law,

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

we take the curl of both sides:

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}).$$

Using the vector identity

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2\vec{E},$$

together with $\nabla \cdot \vec{E} = 0$ and Ampere–Maxwell's law, we obtain

$$-\nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\mu_0 \vec{j} + \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}\right).$$

Rearranging,

$$\nabla^2 \vec{E} - \mu_0 \frac{\partial \vec{j}}{\partial t} - \varepsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0.$$

Substituting the current density from (3), we obtain

$$\boxed{\nabla^2 \vec{E} - \frac{ine^2\mu_0}{m\omega}\frac{\partial \vec{E}}{\partial t} - \varepsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0}.$$

1.5 (5)

For the plane electromagnetic wave

$$\vec{E} = (0, 0, E_0 e^{i(kx-\omega t)}),$$

we have

$$\nabla^2 \vec{E} = -k^2 \vec{E}, \qquad \frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \qquad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}.$$

Substituting these into the wave equation,

$$-k^2 \vec{E} - \frac{ne^2\mu_0}{m}\vec{E} + \omega^2 \varepsilon_0 \mu_0 \vec{E} = 0.$$

Thus,

$$k^2 = \omega^2 \varepsilon_0 \mu_0 - \frac{ne^2\mu_0}{m}.$$

Using $\varepsilon_0 \mu_0 = 1/c^2$, we obtain

$$k^2 = \frac{\omega^2}{c^2}\left(1 - \frac{ne^2}{m\varepsilon_0 \omega^2}\right).$$

Therefore, the dispersion relation is

$$\boxed{k^2 = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)},$$

where the plasma angular frequency is

$$\boxed{\omega_p = \sqrt{\frac{ne^2}{m\varepsilon_0}}}.$$

2 II

2.1 (6)

When $\omega < \omega_p$, the dispersion relation gives

$$k^2 = \frac{\omega^2 - \omega_p^2}{c^2} < 0.$$

Therefore, the wave number must be purely imaginary. We can write

$$k = \pm i\kappa, \qquad \kappa = \frac{\sqrt{\omega_p^2 - \omega^2}}{c} > 0.$$

Assuming that the plasma occupies the region $x > 0$, the electric field must remain finite as $x \to \infty$. Thus, we choose $k = i\kappa$, which gives

$$\begin{aligned}\vec{E} &= E_0 e^{i(kx-\omega t)}\hat{z} \\ &= E_0 e^{-\kappa x} e^{-i\omega t}\hat{z}.\end{aligned}$$

The electric field therefore decays exponentially inside the plasma, indicating that the wave is evanescent rather than propagating.

The characteristic penetration depth is

$$\delta = \frac{1}{\kappa} = \frac{c}{\sqrt{\omega_p^2 - \omega^2}}.$$

For $\omega \ll \omega_p$, we obtain

$$\boxed{\delta \simeq \frac{c}{\omega_p}}.$$

Consequently, the incident electromagnetic wave is reflected at the plasma boundary, while an evanescent field penetrates a finite distance into the plasma.

2.2 (7)

For $\omega \gg \omega_p$, the wave number is

$$k = \frac{\omega}{c}\sqrt{1 - \frac{\omega_p^2}{\omega^2}}.$$

Applying the Taylor expansion $\sqrt{1-x} \simeq 1 - x/2$ for $x \ll 1$,

$$\begin{aligned}k &\simeq \frac{\omega}{c}\left(1 - \frac{\omega_p^2}{2\omega^2}\right) \\ &= \frac{\omega}{c} - \frac{\omega_p^2}{2\omega c}.\end{aligned}$$

Since the pulse is a wave packet, its propagation speed is the group velocity:

$$v_g = \frac{d\omega}{dk} = \left(\frac{dk}{d\omega}\right)^{-1}.$$

Differentiating the approximate expression for k,

$$\frac{dk}{d\omega} \simeq \frac{1}{c} + \frac{\omega_p^2}{2c\omega^2}.$$

Therefore,

$$v_g \simeq \frac{c}{1 + \dfrac{\omega_p^2}{2\omega^2}}.$$

Expanding to second order in ω_p/ω, we finally obtain

$$\boxed{v_g \simeq c\left(1 - \frac{\omega_p^2}{2\omega^2}\right)}.$$

2.3 (8)

The time required for a pulse of angular frequency ω to travel through the plasma of thickness d is

$$t(\omega) = \frac{d}{v_g(\omega)}.$$

Using the approximation from (7),

$$t(\omega) \simeq \frac{d}{c}\left(1 + \frac{\omega_p^2}{2\omega^2}\right).$$

Since the pulse at ω_0 arrives later than the pulse at $2\omega_0$, their arrival-time difference is

$$\Delta T = t(\omega_0) - t(2\omega_0).$$

Substituting the propagation times,

$$\begin{aligned}\Delta T &\simeq \frac{d}{c}\left(1 + \frac{\omega_p^2}{2\omega_0^2}\right) - \frac{d}{c}\left(1 + \frac{\omega_p^2}{8\omega_0^2}\right) \\ &= \frac{3d\omega_p^2}{8c\omega_0^2}.\end{aligned}$$

Since

$$\omega_p^2 = \frac{ne^2}{m\varepsilon_0},$$

we obtain

$$\Delta T = \frac{3nde^2}{8m\varepsilon_0 c\omega_0^2}.$$

Therefore, the electron column density is

$$\boxed{nd = \frac{8m\varepsilon_0 c\omega_0^2 \Delta T}{3e^2}}.$$